Lesson 1 — Extra Practice

Review: Integer Operations and Order of Operations

Part A — Nested Brackets and Multi-Step BEDMAS

1. Evaluate each expression. Work from the **innermost** bracket outwards, and show every step.
(a) $-2[3-(4-7)^2]+5$ (b) $[(-5)+3]^2-[(-4)(2)+6]$ (c) $-3[(-2)^3+10]\div(-6)$

2. Evaluate each expression, showing every step.
(a) $[(-4)^2-(-2)^3]\div(-6+2)$ (b) $20\div(-2)^2-3(1-6)$ (c) $-[4-2(3-8)]$

3. Evaluate each expression.
(a) $(-2)^4-(-2)^3+(-2)^2$ (b) $-1^{10}+(-1)^{10}-(-1)^9$ (c) $(-3)^2(-2)^3\div(-6)^2$

Part B — Where the Brackets Sit: $-a^n$ versus $(-a)^n$

4. Evaluate each expression. Two of the four have the same value — circle them.
(a) -2^4 (b) $(-2)^4$ (c) $-(-2)^4$ (d) $-(-2)^3$

5. Evaluate each expression.
(a) $-3^2\times(-2)^2$ (b) $(-3)^2-3^2$ (c) $(-4)^2\div(-2)^4$ (d) $(-1)^7\times(-2)^3$

6. **Without** evaluating any of them, state whether each expression is positive or negative. Beside each answer, write what decided it.
(a) $(-7)^{11}$ (b) -7^{12} (c) $(-7)^{12}$ (d) $-(-7)^{11}$

Part C — Working Backwards

7. Insert **one** pair of brackets into each expression so that it has the value shown. (Each has exactly one correct answer.)

(a) $3 + 5 \times 2 - 8 \div 4$ so that the value is 14. *(Check: without brackets the value is 11.)*

(b) $10 - 4 \times 3 + 2$ so that the value is -10. *(Check: without brackets the value is 0.)*

8. Replace each $\square$ with one of $+$, $-$, $\times$, or $\div$ to make the statement true. Each operation is used exactly once.

(a) $-12 \square 4 = -3$ (b) $-12 \square 4 = -8$ (c) $-12 \square 4 = -16$ (d) $-12 \square 4 = -48$

9. Find the missing integer in each statement.

(a) $-5 + \square = -12$ (b) $\square \div (-6) = 7$ (c) $(-3) \times \square + 4 = 25$ (d) $\square^2 - 5 = 20$ *(list **all** answers)*

Part D — Error Analysis

10. Nadia evaluated $-3(2-5)^2 + 8 \div (-4)$ as shown below. Exactly one step is the **first** incorrect step. Identify it, explain in one sentence what she did wrong, then give the correct value of the expression.

Step 1: $-3(-3)^2 + 8 \div (-4)$ Step 2: $-3(-9) + 8 \div (-4)$ Step 3: $27 + (-2)$ Step 4: 25

11. Each solution below contains exactly one error. Name the step where it happens, say what rule was broken, and give the correct value.

(a) Evaluate $12 \div (-2) \times 3$.
Step 1: $12 \div (-6)$ Step 2: -2

(b) Evaluate $-4^2 + (-4)^2$.
Step 1: $16 + 16$ Step 2: 32

Part E — Substitution and Integer Reasoning

12. Evaluate each expression when $a = -3$, $b = 4$, and $c = -2$. Substitute using brackets, then apply the order of operations.

(a) $a^2 - 2ab$ (b) $b^2 - 4ac$ (c) $(a + c)^2$ (d) $a^2 + c^2$

Compare your answers to 12(c) and 12(d). What does the comparison tell you about squaring a sum?

13. Evaluate each expression when $x = -2$ and $y = -5$.

(a) x^2y (b) xy^2 (c) $-x^2 - y$ (d) $(xy)^2$

14. State whether each statement is **true** or **false**. If it is false, give a counterexample (one specific set of numbers that breaks it).

(a) The product of three negative integers is always negative.

(b) If the product of two integers is negative, then exactly one of the two integers is negative.

(c) $(-n)^2 = -n^2$ for every integer n.

(d) $-n$ is a negative integer for every integer n.

15. The product of three integers is negative, and none of the three is zero. List every possible combination of signs for the three integers, then complete this sentence: "A product is negative exactly when the number of negative factors is _____."

Part F — Applications and Challenge

For 16 and 17, write **one** integer expression that uses multiplication for the repeated changes, then evaluate it.

16. At midnight a weather station recorded $-3°$C. The temperature then fell $4°$C each hour for the next 5 hours, and after that rose $2°$C each hour for 3 hours. Determine the temperature 8 hours after midnight.

17. Dana's account balance is $-\$45$ (overdrawn). She deposits \$30 each week for 3 weeks, and the bank charges a \$12 overdraft fee on two of those weeks. Determine her balance after the 3 weeks.

18. *(Challenge)* Recall that $|x|$ means the distance of x from zero on a number line, so $|-7| = 7$ and $|5| = 5$. Find **all** integer values of n for which $|n - 2| < 4$. How many values are there?

19. *(Challenge)* The integers -6, 3, and -2 are placed into the three boxes of $\square \times \square + \square$, using each integer exactly once. Determine the **greatest** and the **least** value that can be produced, and state the arrangement that gives each.

Answer Key. 1. (a) 17 (b) 6 (c) 1. **2.** (a) -6 (b) 20 (c) -14. **3.** (a) 28 (b) 1 (c) -2. **4.** (a) -16 (b) 16 (c) -16 (d) 8; (a) and (c) match. **5.** (a) -36 (b) 0 (c) 1 (d) 8. **6.** (a) negative (odd exponent on a negative base) (b) negative (the $-$ is outside, so the even power is made negative) (c) positive (even) (d) positive (odd power is negative, then the outside $-$ flips it). **7.** (a) $(3+5)\times 2-8\div 4=14$ (b) $10-4\times(3+2)=-10$. **8.** (a) $\div$ (b) $+$ (c) $-$ (d) $\times$. **9.** (a) -7 (b) -42 (c) -7 (d) 5 and -5. **10.** Step 2: she wrote $(-3)^2=-9$; squaring a negative gives $+9$. Correct: $-3(9)+8\div(-4)=-27-2=-29$. **11.** (a) Step 1: division and multiplication are done together, left to right, so $12\div(-2)$ comes first; $-6\times 3=-18$. (b) Step 1: $-4^2=-16$ (the exponent applies only to the 4), not 16; $-16+16=0$. **12.** (a) 33 (b) -8 (c) 25 (d) 13; $(a+c)^2\neq a^2+c^2$ — you must add first, then square. **13.** (a) -20 (b) -50 (c) 1 (d) 100. **14.** (a) True. (b) True. (c) False — e.g. $n=3$ gives $(-3)^2=9$ but $-3^2=-9$. (d) False — e.g. $n=-4$ gives $-n=4$, which is positive (and $n=0$ gives 0). **15.** Either all three are negative, or exactly one is negative and the other two are positive; the number of negative factors is *odd*. **16.** $-3+5(-4)+3(2)=-3-20+6=-17$°C. **17.** $-45+3(30)-2(12)=-45+90-24=\$21$. **18.** $-2<n<6$, so $n=-1,0,1,2,3,4,5$ — 7 values. **19.** Greatest $=15$, from $(-6)\times(-2)+3$; least $=-20$, from $(-6)\times 3+(-2)$ (or $3\times(-6)+(-2)$).

Lesson 2 — Extra Practice

Review: Operations with Fractions and Decimals

Part A — Order of Operations with Three or More Terms

1. Evaluate. Remember that multiplication and division come before addition and subtraction. Give each answer in lowest terms.
(a) $\frac{2}{3} + \frac{1}{4} \times \frac{4}{5}$ (b) $\frac{5}{6} - \frac{1}{3} \div \frac{4}{9}$ (c) $\frac{3}{4} \times \frac{8}{9} + \frac{1}{6}$

2. Evaluate. Brackets first.
(a) $\left(\frac{1}{2} + \frac{1}{3}\right) \div \frac{5}{6}$ (b) $\frac{7}{8} - \left(\frac{1}{4} + \frac{1}{3}\right)$ (c) $\frac{2}{5} \times \left(\frac{3}{4} - \frac{1}{2}\right)$

3. Evaluate. These use decimals instead of fractions, but the order of operations does not change.
(a) $0.8 + 0.25 \times 1.2$ (b) $(2.4 - 0.9) \div 0.5$ (c) $1.5 \times 0.4 + 0.6 \div 0.2$

Part B — Mixed Numbers and Complex Fractions

4. Subtract. In each case the fraction being taken away is larger than the fraction it is being taken from, so you will need to **regroup** (borrow 1 from the whole number as a fraction) or convert to improper fractions first. Write each answer as a mixed number in lowest terms.
(a) $5\frac{1}{6} - 2\frac{3}{4}$ (b) $4\frac{1}{3} - 1\frac{5}{6}$ (c) $7\frac{2}{5} - 3\frac{7}{10}$

5. Multiply or divide. Convert every mixed number to an improper fraction first.
(a) $2\frac{1}{2} \times 1\frac{3}{5}$ (b) $3\frac{3}{4} \div 1\frac{1}{2}$ (c) $1\frac{2}{3} \times 2\frac{1}{4}$

6. A **complex fraction** is a fraction whose numerator and denominator are themselves fractions. The main fraction bar still means "divide," so rewrite each one as a division statement and then multiply by the reciprocal.

(a) $\dfrac{\frac{3}{4}}{\frac{2}{3}}$ (b) $\dfrac{\frac{5}{6}}{\frac{10}{9}}$ (c) $\dfrac{2\frac{1}{4}}{\frac{3}{8}}$

Part C — Negative Fractions

7. Add or subtract. The integer sign rules from Lesson 1 apply to the numerators.

(a) $-\frac{3}{4} + \frac{1}{6}$ (b) $\frac{2}{5} - \left(-\frac{1}{3}\right)$ (c) $-\frac{5}{8} - \frac{3}{4}$

8. Multiply or divide. Decide the sign of the answer before you start.

(a) $\left(-\frac{2}{3}\right) \times \left(-\frac{9}{10}\right)$ (b) $-\frac{4}{5} \div \frac{2}{15}$ (c) $\left(-\frac{3}{8}\right) \div \left(-\frac{9}{16}\right)$

9. Evaluate using the correct order of operations.

(a) $-\frac{1}{2} + \left(-\frac{1}{3}\right) \times \frac{3}{5}$ (b) $-\frac{3}{4} - \left(-\frac{1}{2}\right) \div \frac{2}{3}$ (c) $\left(-\frac{2}{5}\right) \times \left(-\frac{5}{6}\right) - \frac{1}{4}$

Part D — Repeating Decimals, Terminating Decimals, and Ordering

10. A bar over a digit means that digit repeats forever, so $0.\overline{6} = 0.666\ldots$ A single repeating digit can be written over 9, and a repeating *pair* of digits over 99. Use this to convert each repeating decimal to a fraction in lowest terms.

(a) $0.\overline{6}$ (b) $0.\overline{4}$ (c) $0.\overline{27}$ (d) $0.\overline{18}$

11. **Without dividing**, state whether each decimal form *terminates* or *repeats*. First simplify the fraction to lowest terms, then look at the prime factors of the denominator. Watch part (d) and part (e) carefully.

(a) $\frac{3}{16}$ (b) $\frac{7}{12}$ (c) $\frac{9}{20}$ (d) $\frac{15}{24}$ (e) $\frac{14}{21}$

12. **Explain** why $\frac{1}{3}$ gives a repeating decimal but $\frac{1}{4}$ gives a terminating one. Your explanation should mention what our number system counts in, and which prime factors a denominator must be built from for the decimal to end.

13. Arrange from **least to greatest**. Converting everything to a decimal is the fastest way to compare.

$$\frac{3}{5}, \qquad 0.58, \qquad \frac{5}{8}, \qquad 0.\overline{6}, \qquad \frac{7}{12}$$

14. Arrange from **least to greatest**. Be careful — with negatives, the number farthest from zero is the *smallest*.

$$-\frac{1}{2}, \qquad -0.45, \qquad -\frac{3}{8}, \qquad -0.6, \qquad -\frac{5}{12}$$

Part E — Working Backwards and Applications

15. Work backwards to find each unknown number. Show the operation you used to undo the one in the question.

(a) $\frac{3}{4}$ of a number is 18. What is the number?

(b) What must be added to $\frac{5}{8}$ to make $1\frac{1}{4}$?

(c) $\frac{2}{5}$ of a number is 14. What is the number?

(d) A number multiplied by $\frac{2}{3}$ gives $1\frac{1}{5}$. Find the number.

16. One loaf of bread uses $3\frac{1}{4}$ cups of flour. Mateo wants to bake $2\frac{1}{2}$ loaves.

(a) How much flour does he need?

(b) He has 8 cups of flour. Does he have enough? If not, how much more does he need?

17. Dana's fuel tank holds 60 L when full. The gauge reads $\frac{3}{8}$ full at the start of a trip and $\frac{1}{6}$ full at the end.

(a) What fraction of a full tank did she use?

(b) How many litres is that?

18. A jacket regularly costs \$84. It is marked down by $\frac{1}{3}$, and then a coupon takes a further $\frac{1}{4}$ off the *sale* price.
(a) What is the final price?
(b) What single fraction of the original price does the customer actually pay?

19. **Stretch.** Find a fraction that is strictly between $\frac{1}{3}$ and $\frac{2}{5}$ — bigger than one, smaller than the other.
(a) Give one such fraction and show it works.
(b) **Explain** the method you used, and why it will always work for any two different fractions.

Part F — Review Challenge

20. *(Multiple Choice)* Which expression has the **greatest** value?
A. $\frac{3}{4} \times \frac{2}{3}$ B. $\frac{3}{4} \div \frac{2}{3}$
C. $\frac{3}{4} - \frac{2}{3}$ D. $\frac{3}{4} + \frac{2}{3}$

21. *(Numerical Response)* Evaluate $\left(\frac{1}{2} + \frac{1}{4}\right) \div \frac{1}{8}$.
The answer is _____.

Answer Key. 1. (a) $\frac{13}{15}$ (b) $\frac{1}{12}$ (c) $\frac{5}{6}$. **2.** (a) 1 (b) $\frac{7}{24}$ (c) $\frac{1}{10}$. **3.** (a) 1.1 (b) 3 (c) 3.6. **4.** (a) $2\frac{5}{12}$ (b) $2\frac{1}{2}$ (c) $3\frac{7}{10}$. **5.** (a) 4 (b) $2\frac{1}{2}$ (c) $3\frac{3}{4}$. **6.** (a) $1\frac{1}{8}$ (b) $\frac{3}{4}$ (c) 6. **7.** (a) $-\frac{7}{12}$ (b) $\frac{11}{15}$ (c) $-1\frac{3}{8}$. **8.** (a) $\frac{3}{5}$ (b) -6 (c) $\frac{2}{3}$. **9.** (a) $-\frac{7}{10}$ (b) 0 (c) $\frac{1}{12}$. **10.** (a) $\frac{2}{3}$ (b) $\frac{4}{9}$ (c) $\frac{3}{11}$ (d) $\frac{2}{11}$. **11.** (a) terminates (b) repeats (c) terminates (d) $\frac{15}{24} = \frac{5}{8}$, terminates (e) $\frac{14}{21} = \frac{2}{3}$, repeats. **12.** We write numbers in base ten, and $10 = 2 \times 5$. A decimal ends only when the denominator in lowest terms is built from 2s and 5s alone, because only then can it be rewritten with a denominator of 10, 100, 1000, ... Here $4 = 2 \times 2$, so $\frac{1}{4} = \frac{25}{100} = 0.25$ terminates; 3 is neither a 2 nor a 5, so no power of ten is a multiple of 3 and $\frac{1}{3} = 0.\overline{3}$ never ends. **13.** $0.58 < \frac{7}{12} < \frac{3}{5} < \frac{5}{8} < 0.\overline{6}$ (i.e. $0.58 < 0.58\overline{3} < 0.6 < 0.625 < 0.\overline{6}$). **14.** $-0.6 < -\frac{1}{2} < -0.45 < -\frac{5}{12} < -\frac{3}{8}$ (i.e. $-0.6 < -0.5 < -0.45 < -0.41\overline{6} < -0.375$). **15.** (a) 24 (b) $\frac{5}{8}$ (c) 35 (d) $1\frac{4}{5}$. **16.** (a) $8\frac{1}{8}$ cups (b) No — he is $\frac{1}{8}$ cup short. **17.** (a) $\frac{5}{24}$ of a tank (b) 12.5 L. **18.** (a) \$42 (b) $\frac{1}{2}$ of the original price. **19.** (a) Many answers; e.g. $\frac{11}{30}$, since $\frac{1}{3} = \frac{10}{30}$ and $\frac{2}{5} = \frac{12}{30}$, so $\frac{10}{30} < \frac{11}{30} < \frac{12}{30}$. ($\frac{3}{8} = 0.375$ also works.) (b) Rewrite both fractions with a common denominator, then build the denominator up further (here 15 to 30) until there is a whole number strictly between the two numerators. Doubling the denominator always doubles the gap between the numerators, so eventually a numerator fits in between — which means there is always another fraction between any two different fractions. **20.** D ($\frac{17}{12}$; A = $\frac{1}{2}$, B = $\frac{9}{8}$, C = $\frac{1}{12}$). **21.** 6.

Lesson 3 — Extra Practice

Prime Factorization, GCF, and LCM

Part A — Bigger Numbers, Deeper Factorizations

1. Write the prime factorization of each number in **exponent form**.
(a) 360 (b) 504 (c) 1000 (d) 1024

2. Use the method indicated. Show every step, then write the answer in exponent form.
(a) Factor tree: 1176 (b) Division ladder: 2310

3. Each prime factorization below belongs to a whole number. Determine that number.
(a) $2^4 \times 3^2 \times 5$ (b) $2^2 \times 5^2 \times 13$ (c) $3^3 \times 7^2$

4. **Counting factors.** Here is a shortcut worth knowing. The number $12 = 2^2 \times 3^1$ has factors $1, 2, 3, 4, 6, 12$ — six of them. Notice that adding 1 to each exponent and multiplying gives $(2+1) \times (1+1) = 6$. That works for every number: take the prime factorization, add 1 to each exponent, multiply. Use the rule to find how many factors each number has.
(a) $2^5 \times 3^2$ (b) 360 (c) 96

5. Work backwards from the ideas above.
(a) What is the smallest whole number that has exactly **three different prime factors**? (b) What is the smallest whole number that has exactly **six factors** in total?

Part B — GCF and LCM with Three Numbers

6. Use prime factorization to determine the greatest common factor.
(a) 84, 126, and 210 (b) 72, 120, and 180 (c) 90, 135, and 225

7. Use prime factorization to determine the lowest common multiple.
(a) 12, 18, and 30 (b) 8, 14, and 35 (c) 15, 24, and 40

8. Determine both the greatest common factor *and* the lowest common multiple of 126 and 168.

9. Write each fraction in lowest terms by dividing the numerator and the denominator by their GCF. State the GCF you used.
(a) $\frac{168}{210}$ (b) $\frac{336}{504}$ (c) $\frac{225}{315}$

Part C — The GCF–LCM Relationship

10. For any two whole numbers a and b, there is a surprising connection:

$$\text{GCF}(a, b) \times \text{LCM}(a, b) = a \times b$$

Verify that the relationship holds for each pair by computing all four quantities.
(a) 18 and 24 (b) 20 and 50

11. Use the relationship from Question 10 to find the missing number. Check your answer by finding the GCF and LCM of the pair you end up with.
(a) Two numbers have a GCF of 6 and an LCM of 72. One of the numbers is 18. What is the other number?
(b) Two numbers have a GCF of 12 and an LCM of 180. One of the numbers is 36. What is the other number?

12. The product of two whole numbers is 3024 and their greatest common factor is 12. Determine their lowest common multiple *without* finding the two numbers.

Part D — Reasoning About Primes, GCF, and LCM

13. For some pairs of numbers, $\text{GCF}(a, b) = a$. Describe exactly when this happens, and give an example of such a pair.

14. For some pairs of numbers, $\text{LCM}(a, b) = a \times b$. Describe exactly when this happens, and give an example of such a pair.

15. **Explain** why 2 is the only even prime number.

16. State whether each statement is **true** or **false**, and justify your answer in one sentence.

(a) If two numbers are both even, then their GCF must be even.

(b) The LCM of two different whole numbers is always greater than both of them.

(c) The GCF of two numbers is always a factor of their LCM.

(d) Every whole number greater than 1 has at least one prime factor.

Part E — Applications and a Stretch

17. A rectangular hallway floor measures 132 cm by 180 cm. It is to be covered exactly by identical square tiles, with no tile cut and no gaps. (a) What is the largest possible side length of a tile? (b) How many tiles of that size are needed?

18. For each situation, first state whether you need the **GCF** or the **LCM**, then solve.

(a) Ferry A leaves the dock every 36 minutes and Ferry B every 48 minutes. They just left together. How long until they next leave together, in hours and minutes?

(b) A teacher has 84 pencils and 126 erasers to sort into identical supply kits, using everything, with as many kits as possible. How many kits, and what goes in each?

19. Storage boxes come in two heights: 15 cm and 18 cm. Ravi builds one stack using only 15 cm boxes and a second stack using only 18 cm boxes, and the two stacks end up exactly the same height. (a) What is the shortest possible height of the stacks? (b) How many boxes are in each stack?

20. *(Stretch)* Determine the smallest whole number greater than 0 that is divisible by **every** whole number from 1 to 6. Then extend your reasoning: what is the smallest whole number divisible by every whole number from 1 to 10?

Answer Key. 1. (a) $360 = 2^3 \cdot 3^2 \cdot 5$ (b) $504 = 2^3 \cdot 3^2 \cdot 7$ (c) $1000 = 2^3 \cdot 5^3$ (d) $1024 = 2^{10}$. **2.** (a) $1176 = 2^3 \cdot 3 \cdot 7^2$ (b) $2310 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11$. **3.** (a) 720 (b) 1300 (c) 1323. **4.** (a) $(5+1)(2+1) = 18$ factors (b) $360 = 2^3 \cdot 3^2 \cdot 5$, so $(4)(3)(2) = 24$ factors (c) $96 = 2^5 \cdot 3$, so $(6)(2) = 12$ factors. **5.** (a) $30 = 2 \cdot 3 \cdot 5$ (b) 12. **6.** (a) 42 (b) 12 (c) 45. **7.** (a) 180 (b) 280 (c) 120. **8.** GCF = 42, LCM = 504. **9.** (a) GCF 42; $\frac{4}{5}$ (b) GCF 168; $\frac{2}{3}$ (c) GCF 45; $\frac{5}{7}$. **10.** (a) GCF 6, LCM 72; $6 \cdot 72 = 432$ and $18 \cdot 24 = 432$ — equal. (b) GCF 10, LCM 100; $10 \cdot 100 = 1000$ and $20 \cdot 50 = 1000$ — equal. **11.** (a) $\frac{6 \cdot 72}{18} = 24$; $\mathrm{GCF}(18, 24) = 6$ and $\mathrm{LCM}(18, 24) = 72$. (b) $\frac{12 \cdot 180}{36} = 60$; $\mathrm{GCF}(36, 60) = 12$ and $\mathrm{LCM}(36, 60) = 180$. **12.** LCM $= 3024 \div 12 = 252$. **13.** Exactly when a is a factor of b (so a divides b); e.g. $a = 6$, $b = 18$ gives GCF = 6. **14.** Exactly when the two numbers share no prime factors (GCF = 1); e.g. 8 and 9 give LCM $= 72 = 8 \cdot 9$. **15.** Every even number has 2 as a factor, so an even number bigger than 2 has factors 1, 2, and itself — more than two factors, so it is composite. Only 2 itself escapes this. **16.** (a) True — 2 is a common factor, so the GCF is a multiple of 2. (b) False — if one number is a factor of the other, the LCM equals the larger one, e.g. $\mathrm{LCM}(6, 12) = 12$. (c) True — every prime in the GCF appears in the LCM to at least that power. (d) True — the number is either prime itself or can be broken down into primes. **17.** (a) $\mathrm{GCF}(132, 180) = 12$, so 12 cm tiles. (b) $11 \times 15 = 165$ tiles. **18.** (a) LCM; $\mathrm{LCM}(36, 48) = 144$ min = 2 h 24 min. (b) GCF; $\mathrm{GCF}(84, 126) = 42$ kits, each with 2 pencils and 3 erasers. **19.** LCM; (a) 90 cm (b) six 15 cm boxes and five 18 cm boxes. **20.** 60; for 1 to 10 it is 2520.

Lesson 4 — Extra Practice

Classifying Numbers in the Real Number System

Part A — Evaluate First, Then Classify

1. Evaluate each expression, then list **every** set it belongs to, using the abbreviations N, W, I, Q, $\overline{Q}$, R.

(a) $\sqrt{49}$ (b) $-\sqrt{81}$ (c) $\dfrac{12}{4}$ (d) $\sqrt{0.25}$

2. Evaluate or simplify each expression as far as you can, then list every set it belongs to. In (a), $\sqrt[3]{-27}$ means the number that multiplies by itself *three* times to give -27.

(a) $\sqrt[3]{-27}$ (b) $\dfrac{\pi}{\pi}$ (c) $0.\overline{45}$ (d) $\sqrt{18}$ (e) $-\dfrac{20}{5}$

3. Complete the table. Evaluate each expression *first*, then place a checkmark in every set the resulting number belongs to.

Number	N	W	I	Q	$\overline{Q}$	R	**Number**	N	W	I	Q	$\overline{Q}$	R
$\sqrt{100}$							2π						
$-\dfrac{18}{6}$							$\sqrt{0.09}$						
$\sqrt{2}\times\sqrt{2}$							$-\sqrt{7}$						
$\pi-\pi$							$\dfrac{22}{7}$						

4. Circle the square roots below that are **rational**, then explain in one sentence the test you used to decide.

$$\sqrt{45},\quad \sqrt{64},\quad \sqrt{121},\quad \sqrt{150},\quad \sqrt{169},\quad \sqrt{200}$$

Part B — Always, Sometimes, Never

5. Complete each statement with *always*, *sometimes*, or *never*. If you answer *sometimes*, give one example that works **and** one that does not.

(a) The sum of two irrational numbers is _____ irrational.

(b) The product of two rational numbers is _____ rational.

(c) An integer is _____ a natural number.

(d) A repeating decimal is _____ rational.

(e) The square root of a natural number is _____ irrational.

(f) The difference of two integers is _____ a whole number.

6. Complete each statement with *always*, *sometimes*, or *never*, and justify your answer with an example or a short reason. Read (b) and (c) very carefully.

(a) A real number that is not rational is _____ irrational.

(b) The product of a rational number and an irrational number is _____ irrational.

(c) A number whose decimal never ends is _____ irrational.

(d) The quotient of two natural numbers is _____ a natural number.

7. A set is **closed** under an operation if combining any two members of the set always gives an answer that is still in that set. Decide *closed* or *not closed* for each, and give a counterexample whenever your answer is *not closed*.

(a) Is the set of integers I closed under division?

(b) Is the set of whole numbers W closed under subtraction?

(c) Is the set of rational numbers Q closed under multiplication?

(d) Is the set of irrational numbers $\overline{Q}$ closed under addition?

8. **Error analysis.** Devon says, "Every decimal number is rational, because a decimal is just a fraction written a different way." Devon is wrong. Explain what is true about his statement, what is false about it, and give one counterexample.

9. Maya notices that $\sqrt{5}$ is irrational, so she expects $\sqrt{5} \times \sqrt{5}$ to be irrational too. Evaluate $\sqrt{5} \times \sqrt{5}$. Is $\sqrt{a} \times \sqrt{a}$ rational for *every* whole number a? Explain.

10. **Stretch.** Can an irrational number be raised to a whole-number power and give a **rational** answer? Either give an example that works, or explain why it is impossible.

Part C — Between-ness, Ordering, and Multiple Choice

11. Answer each. For (a) and (b), evaluate the endpoints first.

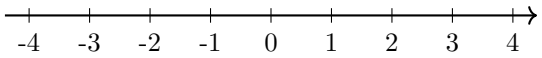
(a) Name one rational number between $\sqrt{16}$ and $\sqrt{25}$.

(b) Name one irrational number between $\sqrt{16}$ and $\sqrt{25}$.

(c) Name one rational number between $\sqrt{2}$ and $\sqrt{3}$.

12. Order the following from **least to greatest**, then plot each one on the number line below. Show the decimal value you used for each.

$$-\frac{7}{2}, \quad \sqrt{10}, \quad \pi, \quad -\sqrt{9}, \quad 2.7, \quad \frac{16}{5}, \quad 0.\overline{3}$$



13. For each description, give one number that fits, **or** write "impossible" and explain in one sentence why no such number exists.

(a) an irrational number that is also an integer

(b) a whole number that is not a natural number

(c) a real number that is neither rational nor irrational

(d) a rational number whose square is irrational

14. *(MC)* Which of the following is **rational**?

A. $\sqrt{30}$

B. $\dfrac{\pi}{2}$

C. $\sqrt{0.16}$

D. $0.1010010001\ldots$

15. *(MC)* The number $\sqrt{2}\times\sqrt{8}$ belongs to exactly the sets

A. $\overline{Q},\ R$

B. $Q,\ R$ only

C. $N,\ W,\ I,\ Q,\ R$

D. $I,\ Q,\ R$ only

16. *(MC)* Which statement is **false**?

A. Every natural number is rational.

B. Every irrational number is real.

C. Every real number is rational.

D. Every integer is real.

Answer Key. 1. (a) 7: N, W, I, Q, R (b) -9: I, Q, R (c) 3: N, W, I, Q, R (d) 0.5: Q, R. **2.** (a) -3: I, Q, R (b) 1: N, W, I, Q, R (c) repeating, so rational $\left(=\frac{5}{11}\right)$: Q, R (d) $\sqrt{18} \approx 4.243$, 18 is not a perfect square: $\overline{Q}, R$ (e) -4: I, Q, R. **3.** $\sqrt{100} = 10$: N, W, I, Q, R; $-\frac{18}{6} = -3$: I, Q, R; $\sqrt{2} \times \sqrt{2} = 2$: N, W, I, Q, R; $\pi - \pi = 0$: W, I, Q, R; $2\pi \approx 6.283$: $\overline{Q}, R$; $\sqrt{0.09} = 0.3$: Q, R; $-\sqrt{7}$: $\overline{Q}, R$; $\frac{22}{7}$: Q, R (it is a fraction of integers, *not* π). **4.** Rational: $\sqrt{64} = 8$, $\sqrt{121} = 11$, $\sqrt{169} = 13$. Irrational: $\sqrt{45}, \sqrt{150}, \sqrt{200}$. Test: $\sqrt{a}$ is rational exactly when the radicand a is a perfect square. **5.** (a) sometimes — $\sqrt{2} + \sqrt{3}$ is irrational, but $\sqrt{2} + (-\sqrt{2}) = 0$ is rational. (b) always — $\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$ is still a fraction of integers. (c) sometimes — 5 is natural, -4 is not. (d) always — every repeating decimal can be written as a fraction. (e) sometimes — $\sqrt{7}$ is irrational, $\sqrt{9} = 3$ is not. (f) sometimes — $5 - 3 = 2$ is whole, $3 - 5 = -2$ is not. **6.** (a) always — R is made up of Q and $\overline{Q}$ only. (b) sometimes — $2 \times \sqrt{2} = 2\sqrt{2}$ is irrational, but $0 \times \sqrt{2} = 0$ is rational. (c) sometimes — π never ends and is irrational, but $\frac{1}{3} = 0.\overline{3}$ never ends and *is* rational (it repeats). (d) sometimes — $6 \div 3 = 2$ is natural, $1 \div 2 = 0.5$ is not. **7.** (a) not closed — $1 \div 2 = 0.5$ is not an integer. (b) not closed — $2 - 5 = -3$ is not a whole number. (c) closed — a product of two fractions of integers is again a fraction of integers. (d) not closed — $\sqrt{2} + (-\sqrt{2}) = 0$, which is rational. **8.** True part: every *terminating* or *repeating* decimal is rational. False part: a decimal that never ends and never repeats is irrational. Counterexample: $\pi = 3.14159\ldots$ (or $0.1010010001\ldots$). **9.** $\sqrt{5} \times \sqrt{5} = 5$, which is rational. Yes — for any whole number a, $\sqrt{a} \times \sqrt{a} = a$, and every whole number is rational, so the product is always rational. **10.** Yes — e.g. $\left(\sqrt{2}\right)^2 = 2$, or $\left(\sqrt{3}\right)^2 = 3$. Both bases are irrational but both answers are rational. **11.** $\sqrt{16} = 4$ and $\sqrt{25} = 5$. (a) e.g. 4.5 or $\frac{9}{2}$ (b) e.g. $\sqrt{17} \approx 4.12$ or $\sqrt{20} \approx 4.47$ (c) $\sqrt{2} \approx 1.41$ and $\sqrt{3} \approx 1.73$, so e.g. 1.5. **12.** $-\frac{7}{2} = -3.5$, $-\sqrt{9} = -3$, $0.\overline{3} \approx 0.33$, 2.7, $\pi \approx 3.142$, $\sqrt{10} \approx 3.162$, $\frac{16}{5} = 3.2$. Order: $-\frac{7}{2} < -\sqrt{9} < 0.\overline{3} < 2.7 < \pi < \sqrt{10} < \frac{16}{5}$. **13.** (a) impossible — every integer is rational, so no integer is irrational. (b) 0. (c) impossible — every real number is either rational or irrational. (d) impossible — the product of two rationals is rational, so a rational squared is rational. **14.** C $\left(\sqrt{0.16} = 0.4\right)$. **15.** C $\left(\sqrt{2} \times \sqrt{8} = \sqrt{16} = 4\right)$. **16.** C (e.g. π is real but not rational).

Lesson 5 — Extra Practice

Perfect Squares, Perfect Cubes, and Estimating Roots

Part A — Testing Big Numbers with Prime Factorization

1. The perfect-square table only reaches 225. For larger numbers, use the prime factorization from Lesson 3: a number is a **perfect square** exactly when *every* exponent in its prime factorization is even. Factor each number, then state whether it is a perfect square. If it is, give its square root.

(a) 1764 (b) 1225 (c) 600 (d) 441

2. A number is a **perfect cube** exactly when every exponent in its prime factorization is divisible by 3. Factor each number, then state whether it is a perfect cube. If it is, give its cube root.

(a) 1728 (b) 2000 (c) 3375 (d) 1080

3. Find the *smallest* whole number each value could be multiplied by so that the product is a perfect square. State the product and its square root.

(a) 12 (b) 18 (c) 75 (d) 500

4. Find the *smallest* whole number each value could be multiplied by so that the product is a perfect cube. State the product and its cube root.

(a) 24 (b) 108 (c) 50

5. Some numbers are a perfect square *and* a perfect cube at the same time.

(a) Show that 64 is both, by giving $\sqrt{64}$ and $\sqrt[3]{64}$. Then write the prime factorization of 64.

(b) Do the same for 729: give $\sqrt{729}$, $\sqrt[3]{729}$, and the prime factorization.

(c) Look at the exponents in your two factorizations. What do they have in common? Use that to explain what makes a number both a perfect square and a perfect cube.

(d) Using your rule from (c), find the next number after 729 that is both a perfect square and a perfect cube.

Part B — Trapping and Estimating Roots

6. State the two consecutive whole numbers each root lies between, and which one it is closer to. Justify by comparing the two distances — not by guessing.
(a) $\sqrt{60}$ (b) $\sqrt{136}$ (c) $\sqrt{1000}$ (d) $\sqrt[3]{500}$

7. Without a calculator, estimate each square root to one decimal place. Write a sentence justifying your estimate, in the style: "$\sqrt{60}$ is between 7 and 8, and nearer to 8 because 60 is closer to 64 than to 49."
(a) $\sqrt{60}$ (b) $\sqrt{136}$ (c) $\sqrt{8}$

8. Squeeze each root between two consecutive *tenths* — for example $4.7 < \sqrt{23} < 4.8$. Do it by squaring candidate tenths, not with the square-root key, and show the two squares you used.
(a) $\sqrt{90}$ (b) $\sqrt{40}$ (c) $\sqrt{23}$ (check the example above by squaring 4.7 and 4.8 yourself)

9. Decide which value in each pair is larger, **without a calculator**. Hint: square both sides (or cube both sides) — a root is easier to compare once the root sign is gone. Show the comparison you used.
(a) $\sqrt{50}$ or 7.1 (b) $\sqrt[3]{30}$ or 3.1 (c) $\sqrt{12}$ or 3.5 (d) $\sqrt{77}$ or 8.8

10. Arrange the following in order from **least to greatest**. Estimate each one first and show your estimates.

$$\sqrt{45}\,, \qquad \sqrt[3]{100}\,, \qquad 6.2\,, \qquad \sqrt{38}\,, \qquad \frac{34}{5}$$

Part C — Working Backwards: Area, Volume, and Surface Area

11. Answer each. Every side length here comes out to a whole number.
(a) A square has an area of 196 cm^2. Find its side length, then find its perimeter.
(b) A cube has a volume of 512 cm^3. Find its edge length, then find its total surface area. (A cube has 6 identical square faces.)

12. Each part runs a formula backwards *twice*. Take it one step at a time.

(a) A cube has a total surface area of 216 m^2. Find the area of one face, then the edge length, then the volume of the cube.

(b) A square has a perimeter of 100 m. Find its side length, then its area. Explain why you did *not* need a square root for this one.

13. A cube-shaped shipping crate has a volume of 1728 cm^3.

(a) Find the edge length of the crate.

(b) Find the total surface area of the crate.

(c) A second crate has *double* the edge length. Find its volume. How many times larger is it than the first crate?

14. A square patio has an area of 150 m^2.

(a) Estimate the side length to one decimal place *without* a calculator, and justify your estimate.

(b) Use a calculator to find the side length to the nearest hundredth of a metre.

(c) Find the perimeter of the patio, to the nearest hundredth of a metre.

Part D — Reasoning and Stretch

15. Jordan claims that $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$ for all whole numbers a and b.

(a) Test the claim using $a = 9$ and $b = 16$. Evaluate both sides and state whether they are equal.

(b) Is Jordan's claim true? Explain what a single counterexample proves about a claim that says "for all".

16. Square roots and cube roots behave differently with negative numbers.

(a) Explain why there is no real number equal to $\sqrt{-16}$. (Think about what happens when you multiply any number by itself.)

(b) Explain why $\sqrt[3]{-64}$ *does* have a real value, and state that value.

(c) Evaluate $\sqrt[3]{-125}$ and $\sqrt[3]{-1000}$.

(d) Complete the rule: the cube root of a negative number is _____, but the square root of a negative number is _____.

17. **Stretch.** Consider all the whole numbers from 1 to 200 inclusive.
(a) How many are perfect squares? List them.
(b) How many are perfect cubes? List them.
(c) How many are *both*? List them, and explain why the count is so small.

18. Answer each.
(a) *(Multiple Choice)* Which of the following is a perfect cube?
A. 196 B. 1764 C. 2744 D. 900
(b) *(Numerical Response)* A cube has a volume of 2744 mm^3. Its edge length, in millimetres, is _____. *(Record your answer from left to right.)*

19. A classmate says: "$\sqrt{1000}$ must be about 31.5, because 1000 is halfway between 961 and 1024." Decide whether the reasoning is sound, then give a better one-decimal estimate with a justification.

Answer Key. 1. (a) $1764 = 2^2 \times 3^2 \times 7^2$ — all exponents even, so yes; $\sqrt{1764} = 42$. (b) $1225 = 5^2 \times 7^2$ — yes; $\sqrt{1225} = 35$. (c) $600 = 2^3 \times 3 \times 5^2$ — the 3 and the 1 are odd, so no. (d) $441 = 3^2 \times 7^2$ — yes; $\sqrt{441} = 21$. **2.** (a) $1728 = 2^6 \times 3^3$ — both exponents divisible by 3, so yes; $\sqrt[3]{1728} = 12$. (b) $2000 = 2^4 \times 5^3$ — 4 is not divisible by 3, so no. (c) $3375 = 3^3 \times 5^3$ — yes; $\sqrt[3]{3375} = 15$. (d) $1080 = 2^3 \times 3^3 \times 5$ — the exponent 1 on 5 is not divisible by 3, so no. **3.** (a) $\times 3 = 36$, $\sqrt{36} = 6$. (b) $\times 2 = 36$, $\sqrt{36} = 6$. (c) $\times 3 = 225$, $\sqrt{225} = 15$. (d) $\times 5 = 2500$, $\sqrt{2500} = 50$. (Multiply by whatever makes every exponent even.) **4.** (a) $24 = 2^3 \times 3$, $\times 9 = 216$, $\sqrt[3]{216} = 6$. (b) $108 = 2^2 \times 3^3$, $\times 2 = 216$, $\sqrt[3]{216} = 6$. (c) $50 = 2 \times 5^2$, $\times 20 = 1000$, $\sqrt[3]{1000} = 10$. **5.** (a) $\sqrt{64} = 8$, $\sqrt[3]{64} = 4$, $64 = 2^6$. (b) $\sqrt{729} = 27$, $\sqrt[3]{729} = 9$, $729 = 3^6$. (c) Every exponent is a multiple of 6 — it is even (so a perfect square) *and* divisible by 3 (so a perfect cube). A number is both exactly when all its prime exponents are multiples of 6. (d) $4^6 = 4096$ (also $4096 = 2^{12}$); $\sqrt{4096} = 64$ and $\sqrt[3]{4096} = 16$. **6.** (a) $49 < 60 < 64$: between 7 and 8, closer to 8 (60 is 11 past 49 but only 4 short of 64). (b) $121 < 136 < 144$: between 11 and 12, closer to 12 (15 past vs. 8 short). (c) $961 < 1000 < 1024$: between 31 and 32, closer to 32 (39 past vs. 24 short). (d) $343 < 500 < 512$: between 7 and 8, closer to 8 (157 past vs. only 12 short). **7.** (a) ≈ 7.7 (calculator 7.746). (b) ≈ 11.7 (calculator 11.662; 11.6 also reasonable). (c) $4 < 8 < 9$, nearer 9, so ≈ 2.8 (calculator 2.828). Justifications will vary in wording. **8.** (a) $9.4^2 = 88.36$ and $9.5^2 = 90.25$, so $9.4 < \sqrt{90} < 9.5$. (b) $6.3^2 = 39.69$ and $6.4^2 = 40.96$, so $6.3 < \sqrt{40} < 6.4$. (c) $4.7^2 = 22.09$ and $4.8^2 = 23.04$, so $4.7 < \sqrt{23} < 4.8$ — the example checks out. **9.** (a) $7.1^2 = 50.41 > 50$, so 7.1 is larger. (b) $3.1^3 = 29.791 < 30$, so $\sqrt[3]{30}$ is larger. (c) $3.5^2 = 12.25 > 12$, so 3.5 is larger. (d) $8.8^2 = 77.44 > 77$, so 8.8 is larger. **10.** $\sqrt[3]{100} \approx 4.6$, $\sqrt{38} \approx 6.2$ (just under), 6.2, $\sqrt{45} \approx 6.7$, $\frac{34}{5} = 6.8$. Order: $\sqrt[3]{100} < \sqrt{38} < 6.2 < \sqrt{45} < \frac{34}{5}$. (Check the close pairs by squaring: $6.2^2 = 38.44 > 38$, and $6.8^2 = 46.24 > 45$.) **11.** (a) side $= \sqrt{196} = 14$ cm; perimeter $= 4 \times 14 = 56$ cm. (b) edge $= \sqrt[3]{512} = 8$ cm; surface area $= 6 \times 8^2 = 384$ cm^2. **12.** (a) one face $= 216 \div 6 = 36$ m^2; edge $= \sqrt{36} = 6$ m; volume $= 6^3 = 216$ m^3. (b) side $= 100 \div 4 = 25$ m; area $= 25^2 = 625$ m^2. No square root is needed — the perimeter gives the side by *dividing*, and area then comes from squaring. **13.** (a) edge $= \sqrt[3]{1728} = 12$ cm. (b) $6 \times 12^2 = 864$ cm^2. (c) edge 24 cm, so volume $= 24^3 = 13\,824$ cm^3 — that is 8 times larger ($2^3 = 8$). **14.** (a) $144 < 150 < 169$, so between 12 and 13, much closer to 12 (6 past 144 vs. 19 short of 169): ≈ 12.2 m. (b) $\sqrt{150} \approx 12.25$ m. (c) $4 \times 12.247\ldots \approx 48.99$ m. **15.** (a) $\sqrt{9+16} = \sqrt{25} = 5$, but $\sqrt{9} + \sqrt{16} = 3 + 4 = 7$. Not equal. (b) The claim is false. One counterexample is enough: "for all" means it must work every time, so a single case where it fails disproves it. **16.** (a) Any real number times itself is positive or zero — a positive times a positive is positive and a negative times a negative is also positive — so nothing squares to -16. (b) Three negatives multiply to a negative: $(-4) \times (-4) \times (-4) = -64$, so $\sqrt[3]{-64} = -4$. (c) $\sqrt[3]{-125} = -5$; $\sqrt[3]{-1000} = -10$. (d) $\ldots$ is negative (a real number); $\ldots$ is not a real number. **17.** (a) 14 of them: $1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196$. (b) 5 of them: $1, 8, 27, 64, 125$. (c) 2 of them: 1 and 64. To be both, every prime exponent must be a multiple of 6 (Q5), which makes such numbers grow very fast — the next one, 729, is already past 200. **18.** (a) C ($2744 = 14^3$; $196 = 14^2$, $1764 = 42^2$ and $900 = 30^2$ are squares, not cubes). (b) 14. **19.** Not sound — 1000 is not halfway: it is 39 above 961 but only 24 below 1024, so it sits about two-thirds of the way up and $\sqrt{1000}$ must be more than 31.5. A better estimate is ≈ 31.6 (calculator 31.62).

Lesson 6 — Extra Practice

Simplifying Square Roots: Entire and Mixed Radicals

Part A — Large Radicands: When the Perfect Square Is Hidden

1. Use prime factorization to convert each entire radical to a mixed radical in **simplest form**. Show the factorization.
(a) $\sqrt{588}$ (b) $\sqrt{1080}$ (c) $\sqrt{2205}$

2. Convert each entire radical to a mixed radical in simplest form.
(a) $\sqrt{1008}$ (b) $\sqrt{845}$ (c) $\sqrt{1183}$

3. Convert each entire radical to a mixed radical in simplest form.
(a) $\sqrt{1250}$ (b) $\sqrt{3200}$ (c) $\sqrt{6125}$

4. Decide whether each entire radical is **already in simplest form**. If it is not, simplify it.
(a) $\sqrt{170}$ (b) $\sqrt{275}$ (c) $\sqrt{286}$ (d) $\sqrt{363}$

Part B — Coefficients, Signs, and Unfinished Work

5. Each radical below already has a coefficient. Simplify the radical part, then multiply the coefficients so the final answer is a mixed radical in simplest form.
(a) $3\sqrt{75}$ (b) $-2\sqrt{98}$ (c) $\frac{1}{2}\sqrt{48}$ (d) $5\sqrt{160}$

6. Each student below pulled out a perfect square that was too small and stopped. Finish the job: write the fully simplified mixed radical.
(a) $\sqrt{192} = 2\sqrt{48}$ (b) $\sqrt{720} = 4\sqrt{45}$ (c) $\sqrt{968} = 2\sqrt{242}$

7. Convert each mixed radical to an **entire radical**. Keep any negative sign outside the root. In parts (c) and (d), simplify the coefficient first.
(a) $10\sqrt{3}$ (b) $-4\sqrt{6}$ (c) $\frac{1}{3}\sqrt{99}$ (d) $\frac{2}{3}\sqrt{45}$

8. Ravi converted $10\sqrt{3}$ to an entire radical and wrote $\sqrt{30}$. Explain his mistake, then give the correct entire radical.

9. Nadia wrote $\sqrt{180} = \sqrt{9 \times 20} = 3\sqrt{20}$. Explain why this is not in simplest form, then finish simplifying it.

Part C — Multiplying and Combining Radicals

The rule $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$ runs in both directions. For example, $\sqrt{6} \times \sqrt{15} = \sqrt{90} = \sqrt{9 \times 10} = 3\sqrt{10}$. Sometimes the product is a whole number with no radical left at all, as in $\sqrt{3} \times \sqrt{27} = \sqrt{81} = 9$.

10. Multiply, then write the result in simplest form.
(a) $\sqrt{6} \times \sqrt{10}$ (b) $\sqrt{2} \times \sqrt{8}$ (c) $\left(2\sqrt{3}\right)^2$ (d) $\sqrt{15} \times \sqrt{35}$

11. Multiply and simplify. Then state which of the three answers is a **rational number** (no radical left).
(a) $3\sqrt{2} \times 4\sqrt{6}$ (b) $\left(5\sqrt{2}\right)^2$ (c) $\sqrt{12} \times \sqrt{27}$

12. **Like radicals.** Two radicals can be added or subtracted only when the number under the root is *the same* — then you simply count them, exactly as with $3x + 5x = 8x$. Often two radicals do not match until you simplify them. For example,

$$\sqrt{8} + \sqrt{32} \;=\; 2\sqrt{2} + 4\sqrt{2} \;=\; 6\sqrt{2}.$$

Simplify each radical first, then combine.
(a) $\sqrt{12} + \sqrt{27}$ (b) $\sqrt{50} - \sqrt{18}$ (c) $\sqrt{45} + \sqrt{80}$ (d) $\sqrt{8} + \sqrt{18} - \sqrt{32}$

13. Simplify each radical, then combine.
(a) $\sqrt{98} - \sqrt{8} + \sqrt{200}$ (b) $2\sqrt{27} + \sqrt{48}$ (c) $\sqrt{24} + \sqrt{54}$

14. Is it true that $\sqrt{a}+\sqrt{b}=\sqrt{a+b}$? Test it with $a=9$ and $b=16$, showing both sides, and state your conclusion.

15. Explain why $\sqrt{2}+\sqrt{3}$ cannot be written as a single mixed radical, but $\sqrt{2}+\sqrt{12}$ *almost* can — and say what goes wrong in the second one too.

Part D — Ordering, Geometry, and Reasoning

16. Without a calculator, arrange the following from **least to greatest** by converting each to an entire radical first: $5\sqrt{3}$, $3\sqrt{6}$, $2\sqrt{14}$, $\sqrt{70}$, $4\sqrt{5}$.

17. Without a calculator, arrange the following from **greatest to least**: $-2\sqrt{5}$, $-3\sqrt{2}$, $-\sqrt{17}$. Explain in one sentence why the ordering reverses once the radicals are negative.

18. A square has an area of 75 square centimetres. Determine its exact side length in simplest mixed radical form, then its exact perimeter.

19. A rectangle measures 6 cm by 9 cm. Use the Pythagorean theorem to find the exact length of its diagonal in simplest mixed radical form.

20. A square tarp has an area of 288 square feet. Determine the exact side length and the exact perimeter, both in simplest mixed radical form.

21. A right triangle has legs of 4 m and 10 m. Determine the exact length of the hypotenuse in simplest mixed radical form.

22. **Stretch.** An entire radical $\sqrt{n}$ can be rewritten as a mixed radical exactly when n has a perfect-square factor greater than 1.
(a) List every whole number n from 2 to 50 for which $\sqrt{n}$ simplifies. *Hint:* run through the multiples of 4, then of 9, then of 25, then of 49, and do not count a number twice.

(b) How many numbers are on your list? Of those, how many give a whole-number answer with no radical left? Which numbers are they?

Answer Key. 1. (a) $588 = 2^2 \cdot 3 \cdot 7^2$, $14\sqrt{3}$ (b) $1080 = 2^3 \cdot 3^3 \cdot 5$, $6\sqrt{30}$ (c) $2205 = 3^2 \cdot 5 \cdot 7^2$, $21\sqrt{5}$. **2.** (a) $12\sqrt{7}$ (b) $13\sqrt{5}$ (c) $13\sqrt{7}$. **3.** (a) $25\sqrt{2}$ (b) $40\sqrt{2}$ (c) $35\sqrt{5}$. **4.** (a) simplest ($170 = 2 \cdot 5 \cdot 17$) (b) $5\sqrt{11}$ (c) simplest ($286 = 2 \cdot 11 \cdot 13$) (d) $11\sqrt{3}$. **5.** (a) $15\sqrt{3}$ (b) $-14\sqrt{2}$ (c) $2\sqrt{3}$ (d) $20\sqrt{10}$. **6.** (a) $8\sqrt{3}$ (b) $12\sqrt{5}$ (c) $22\sqrt{2}$. **7.** (a) $\sqrt{300}$ (b) $-\sqrt{96}$ (c) $\sqrt{11}$ (d) $\sqrt{20}$. **8.** He multiplied 10 by 3 instead of squaring the 10 first; correctly $10\sqrt{3} = \sqrt{100 \times 3} = \sqrt{300}$. **9.** $20 = 4 \times 5$ still holds a perfect-square factor, so $3\sqrt{20}$ is not finished: $3\sqrt{20} = 3 \cdot 2\sqrt{5} = 6\sqrt{5}$, and $\sqrt{180} = 6\sqrt{5}$. **10.** (a) $\sqrt{60} = 2\sqrt{15}$ (b) $\sqrt{16} = 4$ (c) $4 \cdot 3 = 12$ (d) $\sqrt{525} = 5\sqrt{21}$. **11.** (a) $12\sqrt{12} = 24\sqrt{3}$ (b) $25 \cdot 2 = 50$ (c) $\sqrt{324} = 18$; (b) and (c) are rational, (a) is not. **12.** (a) $2\sqrt{3} + 3\sqrt{3} = 5\sqrt{3}$ (b) $5\sqrt{2} - 3\sqrt{2} = 2\sqrt{2}$ (c) $3\sqrt{5} + 4\sqrt{5} = 7\sqrt{5}$ (d) $2\sqrt{2} + 3\sqrt{2} - 4\sqrt{2} = \sqrt{2}$. **13.** (a) $7\sqrt{2} - 2\sqrt{2} + 10\sqrt{2} = 15\sqrt{2}$ (b) $6\sqrt{3} + 4\sqrt{3} = 10\sqrt{3}$ (c) $2\sqrt{6} + 3\sqrt{6} = 5\sqrt{6}$. **14.** No. $\sqrt{9} + \sqrt{16} = 3 + 4 = 7$ but $\sqrt{9 + 16} = \sqrt{25} = 5$, and $7 \neq 5$; the rule is false (it works for multiplication, not addition). **15.** 2 and 3 are different radicands and neither simplifies, so there is nothing to count — $\sqrt{2} + \sqrt{3}$ must stay as it is. For $\sqrt{2} + \sqrt{12}$, simplifying gives $\sqrt{2} + 2\sqrt{3}$; the radicands 2 and 3 are still different, so it also cannot be combined. **16.** Entire forms: $\sqrt{75}$, $\sqrt{54}$, $\sqrt{56}$, $\sqrt{70}$, $\sqrt{80}$, so $3\sqrt{6} < 2\sqrt{14} < \sqrt{70} < 5\sqrt{3} < 4\sqrt{5}$. **17.** Entire forms: $-\sqrt{20}$, $-\sqrt{18}$, $-\sqrt{17}$, so $-\sqrt{17} > -3\sqrt{2} > -2\sqrt{5}$. The bigger the radicand, the bigger the root — but taking the opposite of a bigger number gives a smaller number, so the order flips. **18.** Side $= \sqrt{75} = 5\sqrt{3}$ cm; perimeter $= 20\sqrt{3}$ cm. **19.** $\sqrt{6^2 + 9^2} = \sqrt{117} = 3\sqrt{13}$ cm. **20.** Side $= \sqrt{288} = 12\sqrt{2}$ ft; perimeter $= 48\sqrt{2}$ ft. **21.** $\sqrt{4^2 + 10^2} = \sqrt{116} = 2\sqrt{29}$ m. **22.** (a) 4, 8, 9, 12, 16, 18, 20, 24, 25, 27, 28, 32, 36, 40, 44, 45, 48, 49, 50. (b) 19 numbers; 6 of them are perfect squares and leave no radical — 4, 9, 16, 25, 36, 49.

Lesson 7 — Extra Practice

Simplifying Higher-Index Radicals

Handy beyond the lesson's lists. Perfect cubes: $1728 = 12^3$, $1000 = 10^3$, $343 = 7^3$. Perfect fourth powers: $1296 = 6^4$, $2401 = 7^4$, $4096 = 8^4$. Perfect fifth powers: $3125 = 5^5$, $7776 = 6^5$.

Part A — Hunting for the Perfect Power

1. The radicands below are too large to guess at. Write each one as a product of primes first, then convert it to a mixed radical in simplest form.
(a) $\sqrt[3]{1080}$ (b) $\sqrt[3]{-2000}$ (c) $\sqrt[3]{1372}$

2. Use prime factorization again, this time hunting for a perfect *fourth* power. With index 4, a prime must appear **four** times before one copy may leave the root sign.
(a) $\sqrt[4]{1250}$ (b) $\sqrt[4]{2000}$ (c) $\sqrt[4]{3888}$

3. Now index 5 — a prime must appear **five** times to escape. One of these is already in simplest form; say which, and say how the prime factorization told you so.
(a) $\sqrt[5]{486}$ (b) $\sqrt[5]{-1944}$ (c) $\sqrt[5]{2500}$

4. Each of these is a perfect power, so no radical sign should be left in your answer.
(a) $\sqrt[4]{625}$ (b) $\sqrt[5]{243}$ (c) $\sqrt[3]{0.008}$ (d) $\sqrt[4]{4096}$ (e) $\sqrt[3]{-0.064}$

Part B — Mixed and Entire at Higher Indices

5. Convert each mixed radical to an entire radical. Raise the coefficient to the power of the *index*, not always to the power 3.
(a) $2\sqrt[3]{5}$ (b) $3\sqrt[4]{2}$ (c) $2\sqrt[5]{3}$ (d) $5\sqrt[3]{4}$

6. Convert each mixed radical to an entire radical. Every index here is **odd**, so the negative coefficient may be absorbed — but track the sign of the radicand carefully.
(a) $-2\sqrt[3]{3}$ (b) $-3\sqrt[3]{2}$ (c) $-2\sqrt[5]{2}$ (d) $-4\sqrt[3]{2}$

7. Jonah tries to write $-2\sqrt[4]{3}$ as a single entire radical. He computes $(-2)^4 = 16$ and writes $-2\sqrt[4]{3} = \sqrt[4]{48}$. Evaluate both expressions on a calculator to show he is wrong, explain what raising a negative coefficient to an *even* power did to the sign, and write the correct entire form.

8. (a) Convert $2\sqrt[3]{5}$ and $3\sqrt[3]{2}$ to entire radicals.
(b) Use the two entire radicals to decide which of the original mixed radicals is larger. Explain why comparing entire radicals is easier than comparing mixed ones.

9. Convert $\sqrt[3]{-1024}$ to a mixed radical in simplest form, then convert your answer back to an entire radical. You should arrive at $\sqrt[3]{-1024}$ again — show that you do.

Part C — Odd Index, Even Index

10. Sort the eight radicals below into the two columns. For each one that *is* real, write its exact value in simplest form; for each one that is not, write "not a real number."

$$\sqrt[3]{-64},\quad \sqrt[4]{-81},\quad \sqrt{-25},\quad \sqrt[5]{-32},\quad \sqrt[6]{-1},\quad \sqrt[3]{-125},\quad \sqrt[4]{16},\quad \sqrt[7]{-128}$$

Real — give the exact value **Not a real number**

11. Marcus says the index only affects *how* you find a root, never *whether* a root exists. Use $\sqrt[3]{-8}$ and $\sqrt[4]{-16}$ to show he is wrong. For each one, write the multiplication check you would do ($_ \times _ \times \cdots =$ the radicand) and say what that check reveals.

12. (a) For which whole-number indices n is $\sqrt[n]{-32}$ a real number? State the rule, not just a list.
(b) Evaluate $\sqrt[5]{-32}$, then write $\sqrt[3]{-32}$ as a mixed radical in simplest form.

13. Evaluate each of $\sqrt{(-6)^2}$, $\sqrt[3]{(-6)^3}$, $\sqrt[4]{(-6)^4}$ and $\sqrt[5]{(-6)^5}$. Then complete: $\sqrt[n]{a^n} = a$ when the index is _____, but equals the *positive* version of a when the index is _____.

Part D — Comparing, Geometry, and a Stretch

14. Which is larger, $\sqrt[3]{20}$ or $\sqrt{8}$? You cannot compare radicands directly when the indices differ. Instead raise *both* radicals to the power 6 — the smallest power that clears an index of 3 and an index of 2 — and compare the whole numbers you get.

15. Arrange $\sqrt{10}$, $\sqrt[3]{30}$ and $\sqrt[4]{100}$ from **least to greatest** by raising all three to the power 12. One surprise is waiting for you — describe it.

16. A cube has a volume of 250 cm^3. Determine its exact edge length in simplest mixed radical form, then state the edge length to the nearest tenth of a centimetre (use a calculator).

17. A second cube has a volume of 375 cm^3. Find its exact edge length in simplest mixed radical form, and state which of the two cubes has the longer edge.

18. **Stretch.** Answer each part, and show the reasoning that got you there rather than guessing and checking forever.

(a) Find the smallest whole number $k > 1$ for which $\sqrt[3]{k}$ is a whole number.

(b) Find the smallest whole number $k > 1$ that is a perfect square *and* a perfect cube.

(c) Find the smallest whole number $k > 1$ that is a perfect square, a perfect cube *and* a perfect fourth power.

19. *(Multiple Choice)* Exactly one of these is **not** in simplest form. Which one?

A. $3\sqrt[3]{6}$ B. $2\sqrt[4]{10}$ C. $2\sqrt[3]{24}$ D. $4\sqrt[5]{7}$

Then write that one in simplest form.

20. *(Numerical Response)* The mixed radical $2\sqrt[5]{7}$ can be written as the entire radical $\sqrt[5]{n}$. The value of n is _____.

Answer Key. 1. (a) $1080 = 2^3 \times 3^3 \times 5$, so $6\sqrt[3]{5}$ (b) $2000 = 2^4 \times 5^3$, so $-10\sqrt[3]{2}$ (c) $1372 = 2^2 \times 7^3$, so $7\sqrt[3]{4}$. **2.** (a) $1250 = 2 \times 5^4$, so $5\sqrt[4]{2}$ (b) $2000 = 2^4 \times 5^3$, so $2\sqrt[4]{125}$ (c) $3888 = 2^4 \times 3^5$, so $6\sqrt[4]{3}$. **3.** (a) $486 = 2 \times 3^5$, so $3\sqrt[5]{2}$ (b) $1944 = 2^3 \times 3^5$, so $-3\sqrt[5]{8}$ (c) already simplest: $2500 = 2^2 \times 5^4$ and no prime appears five or more times. **4.** (a) 5 (b) 3 (c) 0.2 (d) 8 (e) -0.4. **5.** (a) $\sqrt[3]{40}$ (b) $\sqrt[4]{162}$ (c) $\sqrt[5]{96}$ (d) $\sqrt[3]{500}$. **6.** (a) $\sqrt[3]{-24}$ (b) $\sqrt[3]{-54}$ (c) $\sqrt[5]{-64}$ (d) $\sqrt[3]{-128}$. **7.** $-2\sqrt[4]{3} \approx -2.632$ but $\sqrt[4]{48} \approx +2.632$. Raising -2 to the even power 4 destroyed the negative sign, so the coefficient cannot be absorbed. Correct form: $-2\sqrt[4]{3} = -\sqrt[4]{48}$, with the negative left outside the radical. **8.** (a) $2\sqrt[3]{5} = \sqrt[3]{40}$ and $3\sqrt[3]{2} = \sqrt[3]{54}$. (b) $54 > 40$, so $3\sqrt[3]{2}$ is larger (≈ 3.780 vs ≈ 3.420). Entire radicals carry all the size in one number, so with the same index the larger radicand is the larger radical. **9.** $-1024 = -512 \times 2$, so $\sqrt[3]{-1024} = -8\sqrt[3]{2}$; back: $(-8)^3 = -512$ and $-512 \times 2 = -1024$, giving $\sqrt[3]{-1024}$. **10.** Real: $\sqrt[3]{-64} = -4$, $\sqrt[5]{-32} = -2$, $\sqrt[3]{-125} = -5$, $\sqrt[4]{16} = 2$, $\sqrt[7]{-128} = -2$. Not a real number: $\sqrt[4]{-81}$, $\sqrt{-25}$, $\sqrt[6]{-1}$ (each has an even index with a negative radicand). **11.** $\sqrt[3]{-8}$: $(-2) \times (-2) \times (-2) = -8$, so the value is -2 — an odd number of negative factors leaves a negative product. $\sqrt[4]{-16}$: any candidate must satisfy $_ \times _ \times _ \times _ = -16$, but an even number of equal factors is never negative ($2^4 = 16$ and $(-2)^4 = 16$), so there is no real answer. The index decides whether a root exists at all. **12.** (a) Real for every *odd* index n; not real for every even n, since no real number raised to an even power is negative. (b) $\sqrt[5]{-32} = -2$; $\sqrt[3]{-32} = \sqrt[3]{-8 \times 4} = -2\sqrt[3]{4}$. **13.** $\sqrt{(-6)^2} = \sqrt{36} = 6$, $\sqrt[3]{(-6)^3} = \sqrt[3]{-216} = -6$, $\sqrt[4]{(-6)^4} = \sqrt[4]{1296} = 6$, $\sqrt[5]{(-6)^5} = \sqrt[5]{-7776} = -6$. So $\sqrt[n]{a^n} = a$ when n is *odd*, but equals the positive version of a (that is, $|a|$) when n is *even*. **14.** $\left(\sqrt[3]{20}\right)^6 = 20^2 = 400$ and $\left(\sqrt{8}\right)^6 = 8^3 = 512$. Since $512 > 400$, $\sqrt{8}$ is larger (≈ 2.828 vs ≈ 2.714). **15.** $\left(\sqrt{10}\right)^{12} = 10^6 = 1\,000\,000$, $\left(\sqrt[3]{30}\right)^{12} = 30^4 = 810\,000$, $\left(\sqrt[4]{100}\right)^{12} = 100^3 = 1\,000\,000$. So $\sqrt[3]{30} < \sqrt{10} = \sqrt[4]{100}$. The surprise: $\sqrt[4]{100}$ and $\sqrt{10}$ are the *same* number (≈ 3.162), because $\sqrt[4]{100} = \sqrt{\sqrt{100}} = \sqrt{10}$. **16.** Edge $= \sqrt[3]{250} = \sqrt[3]{125 \times 2} = 5\sqrt[3]{2}$ cm ≈ 6.3 cm. **17.** Edge $= \sqrt[3]{375} = \sqrt[3]{125 \times 3} = 5\sqrt[3]{3}$ cm ≈ 7.2 cm; the 375 cm^3 cube has the longer edge. **18.** (a) $k = 8$ (the smallest cube above 1^3 is 2^3). (b) $k = 64 = 2^6$ — k must be a perfect 6th power, since 6 is the smallest number both 2 and 3 divide into. (c) $k = 4096 = 2^{12}$ — now k must be a perfect 12th power, and 12 is the smallest number divisible by 2, 3 and 4. **19.** C. $\sqrt[3]{24} = \sqrt[3]{8 \times 3} = 2\sqrt[3]{3}$, so $2\sqrt[3]{24} = 4\sqrt[3]{3}$. **20.** 224 (since $2^5 \times 7 = 32 \times 7 = 224$).

Lesson 8 — Extra Practice

Unit Review: Number Sense

Every question below pulls together two or more lessons from this unit — exactly what a unit test asks you to do. Work without a calculator unless a question says otherwise.

Part A — Multi-Step Evaluation

1. Evaluate each expression. Watch the order of operations, and remember that $-n^2 \neq (-n)^2$.

(a) $\sqrt{16} + \frac{2}{3} \times 9 - (-2)^3$ (b) $(-3)^2 - \sqrt{49} \times 2$ (c) $\sqrt[3]{-64} + (-5)(-3)$ (d) $-4^2 + \sqrt{81} \div (-3)$

2. Evaluate each expression. Give every answer as an exact fraction or whole number.

(a) $\frac{3}{4} + \frac{1}{6} \times \sqrt{36}$ (b) $\left(\frac{2}{3} - \frac{1}{4}\right) \times \sqrt{144}$ (c) $\sqrt{25} \times \frac{3}{5} - \left(\frac{1}{2}\right)^2$

3. Each sum needs a three-way common denominator. Use the **LCM** of the three denominators as the LCD, state it, then evaluate. Give each answer in lowest terms.

(a) $\frac{1}{6} + \frac{3}{8} + \frac{5}{12}$ (b) $\frac{5}{9} - \frac{1}{6} + \frac{2}{15}$ (c) $\frac{7}{10} + \frac{2}{15} - \frac{1}{4}$

4. Reduce each fraction to lowest terms in *one* step by first finding the **GCF** of the numerator and denominator from their prime factorizations.

(a) $\frac{168}{420}$ (b) $\frac{270}{756}$

Part B — Combining Skills

5. These radicands are too large to factor by inspection. Write the prime factorization of each radicand first, then simplify to a mixed radical in simplest form.

(a) $\sqrt{1512}$ (b) $\sqrt{2352}$ (c) $\sqrt[3]{1080}$ (d) $\sqrt[4]{1250}$

6. Work backwards from a prime factorization. In each case, state the number, decide whether it is a perfect square or a perfect cube, and then give the exact root asked for.

(a) A number's prime factorization is $2^4 \times 3^2$. Is it a perfect square? Give its square root.

(b) A number's prime factorization is $2^3 \times 5^3$. Is it a perfect cube? Give its cube root.

(c) A number's prime factorization is $2^3 \times 3^4$. Is it a perfect square? Simplify its square root anyway.

7. Simplify each expression to its exact value first. Then state whether the *result* is rational or irrational, and list every set it belongs to: N, W, I, Q, $\overline{Q}$, R.

(a) $\sqrt{18} \times \sqrt{2}$ (b) $\dfrac{\sqrt{12}}{\sqrt{3}}$ (c) $\sqrt{27}$ (d) $\sqrt[3]{-8} \times \frac{3}{4}$

8. **Estimate, then give the exact form.** For each root: state the two consecutive whole numbers it lies between, state which one it is closer to, give a one-decimal estimate, and then write the root exactly as a mixed radical in simplest form.

(a) $\sqrt{90}$ (b) $\sqrt{147}$ (c) $\sqrt{200}$

9. Without a calculator, arrange these seven numbers in order from **least to greatest**. Beside your list, show the comparison that settles 3.4, $2\sqrt{3}$ and $\frac{7}{2}$.

$-\frac{5}{2}$, $\sqrt[3]{-27}$, $2\sqrt{3}$, 3.4, $\sqrt{8}$, $\frac{7}{2}$, $\sqrt[3]{64}$

Part C — Reasoning and Proof-Style

10. **Error analysis.** Each solution below has exactly *one* wrong step. Name the step, explain the mistake in one sentence, and give the correct final answer.

(a) Evaluate $\frac{3}{4} + \frac{1}{2} \times (-2)^2 - \sqrt{9}$.
Step 1: $= \frac{3}{4} + \frac{1}{2} \times (-4) - 3$
Step 2: $= \frac{3}{4} - 2 - 3$
Step 3: $= -\frac{17}{4}$

(b) Simplify $\sqrt{720}$.
Step 1: $720 = 2^4 \times 3^2 \times 5$
Step 2: $\sqrt{720} = 2^2 \times 3^2 \times \sqrt{5}$
Step 3: $= 4 \times 9\sqrt{5} = 36\sqrt{5}$

(c) Find the LCM of 18 and 24.
Step 1: $18 = 2 \times 3^2$, $24 = 2^3 \times 3$
Step 2: LCM $= 2 \times 3$
Step 3: LCM $= 6$

11. **Always, sometimes, or never?** For each statement, write always, sometimes or never, and justify with a reason or a counterexample.

(a) The product of two irrational numbers is irrational.

(b) The sum of an integer and a rational number is rational.

(c) If n is a whole number, then $\sqrt{n}$ is irrational.

(d) The LCM of two whole numbers is greater than their GCF.

(e) If n is negative, then $\sqrt[3]{n}$ is a real number.

(f) If n is negative, then $\sqrt{n}$ is a real number.

12. Explain each of the following in complete sentences.

(a) Explain why $\sqrt{2}\times\sqrt{50}$ is a whole number but $\sqrt{2}\times\sqrt{5}$ is not.

(b) A whole number is divisible by both 6 and 8. Explain why it must also be divisible by 24, but need not be divisible by 48.

Part D — Applications

13. A square tabletop has an area of 288 cm^2.

(a) Give the exact side length in simplest mixed radical form. (b) Between which two consecutive whole numbers does the side length lie, and what is it to the nearest tenth of a centimetre (use a calculator)? (c) Give the exact perimeter in simplest mixed radical form.

14. A cube has a volume of 375 cm^3.

(a) Give the exact edge length as a mixed radical in simplest form. (b) Is that edge length rational or irrational? Explain. (c) Between which two consecutive whole numbers does the edge length lie?

15. A rectangular wall measuring 96 cm by 144 cm is covered with the largest possible identical square tiles, with no leftover strip along either edge.
(a) Use the GCF to find the side length of each tile. (b) How many tiles are used in total? (c) If $\frac{1}{3}$ of the tiles are blue and $\frac{1}{2}$ are white, what fraction of the tiles are neither colour — and how many tiles is that?

16. One square garden plot has an area of 200 m^2; a second has an area of 50 m^2.
(a) Give both side lengths exactly, in simplest mixed radical form. (b) How many times longer is the first side than the second? (c) Is that ratio rational or irrational? Explain how two irrational lengths can have a rational ratio.

Part E — Challenge

17. Use prime factorization to answer each. (Hint: a perfect square needs *even* exponents on every prime; a perfect cube needs exponents that are multiples of 3.)
(a) What is the smallest whole number you can multiply 72 by to make a perfect square? (b) What is the smallest whole number you can multiply 72 by to make a perfect cube? (c) What is the smallest whole number you can multiply 40 by to make a perfect square?

18. Find every whole number n that fits both conditions.
(a) $\sqrt{n}$ is between 7 and 8, and n is a multiple of 5. (b) $\sqrt[3]{n}$ is between 4 and 5, and n is a multiple of 9.

19. Work backwards from the simplified form.
(a) $\sqrt{a} = 6\sqrt{2}$. Find a. (b) $\sqrt[3]{b} = 2\sqrt[3]{5}$. Find b. (c) A number's prime factorization is $2^6 \times 3^4 \times 5^2$. Explain why it is a perfect square without multiplying it out, then give its square root as a product of primes.

Answer Key. 1. (a) 18 (b) -5 (c) 11 (d) -19. **2.** (a) $\frac{7}{4}$ (b) 5 (c) $\frac{11}{4}$. **3.** (a) LCD $= 24$; $\frac{23}{24}$ (b) LCD $= 90$; $\frac{47}{90}$ (c) LCD $= 60$; $\frac{35}{60} = \frac{7}{12}$. **4.** (a) GCF $= 84$; $\frac{2}{5}$ (b) GCF $= 54$; $\frac{5}{14}$. **5.** (a) $1512 = 2^3 \times 3^3 \times 7$, $6\sqrt{42}$ (b) $2352 = 2^4 \times 3 \times 7^2$, $28\sqrt{3}$ (c) $1080 = 2^3 \times 3^3 \times 5$, $6\sqrt[3]{5}$ (d) $1250 = 2 \times 5^4$, $5\sqrt[4]{2}$. **6.** (a) 144; yes, both exponents are even; $\sqrt{144} = 12$ (b) 1000; yes, both exponents are multiples of 3; $\sqrt[3]{1000} = 10$ (c) 648; no, the exponent on 2 is odd; $\sqrt{648} = 18\sqrt{2}$. **7.** (a) $\sqrt{36} = 6$; rational; N, W, I, Q, R (b) $\sqrt{4} = 2$; rational; N, W, I, Q, R (c) $\sqrt{27} = 3\sqrt{3}$; irrational; $\overline{Q}, R$ (d) $-2 \times \frac{3}{4} = -\frac{3}{2}$; rational; Q, R. **8.** (a) between 9 and 10, closer to 9 ($90 - 81 = 9$ vs. $100 - 90 = 10$), ≈ 9.5; exactly $3\sqrt{10}$ (b) between 12 and 13, closer to 12, ≈ 12.1; exactly $7\sqrt{3}$ (c) between 14 and 15, closer to 14, ≈ 14.1; exactly $10\sqrt{2}$. **9.** $\sqrt[3]{-27} < -\frac{5}{2} < \sqrt{8} < 3.4 < 2\sqrt{3} < \frac{7}{2} < \sqrt[3]{64}$, i.e. $-3 < -2.5 < 2.83 < 3.4 < 3.46 < 3.5 < 4$. Comparison: $2\sqrt{3} = \sqrt{12}$, and $3.4^2 = 11.56 < 12$ while $\left(\frac{7}{2}\right)^2 = 12.25 > 12$, so $3.4 < 2\sqrt{3} < \frac{7}{2}$. **10.** (a) Step 1 — $(-2)^2 = +4$, not -4 (the brackets square the sign); $\frac{3}{4} + 2 - 3 = -\frac{1}{4}$. (b) Step 2 — taking a square root *halves* each even exponent, so 3^2 becomes 3, not 3^2; $\sqrt{720} = 2^2 \times 3 \times \sqrt{5} = 12\sqrt{5}$. (c) Step 2 — that is the GCF rule; the LCM takes every prime at its *highest* power, so LCM $= 2^3 \times 3^2 = 72$ (and 6 is the GCF). **11.** (a) sometimes — $\sqrt{2} \times \sqrt{2} = 2$ is rational, but $\sqrt{2} \times \sqrt{3} = \sqrt{6}$ is irrational. (b) always — every integer is rational, and a sum of two rationals is rational. (c) sometimes — $\sqrt{7}$ is irrational, but $\sqrt{9} = 3$ is not. (d) sometimes — normally yes, but if the two numbers are equal (e.g. 6 and 6) the LCM and GCF are both 6. (e) always — an odd index can absorb a negative, e.g. $\sqrt[3]{-27} = -3$. (f) never — no real number squares to a negative. **12.** (a) $\sqrt{2} \times \sqrt{50} = \sqrt{100} = 10$ because 100 is a perfect square, while $\sqrt{2} \times \sqrt{5} = \sqrt{10}$ and 10 has no perfect-square factor besides 1, so it stays irrational. (b) Its prime factorization must contain every prime in $6 = 2 \times 3$ and in $8 = 2^3$, so it contains $2^3 \times 3 = 24$, the LCM; but $48 = 2^4 \times 3$ needs a fourth 2 that neither 6 nor 8 forces — e.g. 24 itself is divisible by 6 and 8 but not by 48. **13.** (a) $\sqrt{288} = 12\sqrt{2}$ cm (b) between 16 and 17, closer to 17; ≈ 17.0 cm (c) $4 \times 12\sqrt{2} = 48\sqrt{2}$ cm. **14.** (a) $\sqrt[3]{375} = 5\sqrt[3]{3}$ cm (b) irrational — $375 = 3 \times 5^3$ is not a perfect cube, so the $\sqrt[3]{3}$ never ends or repeats (c) between 7 and 8, since $7^3 = 343$ and $8^3 = 512$. **15.** (a) GCF$(96, 144) = 48$, so 48 cm (b) $2 \times 3 = 6$ tiles (c) $1 - \frac{1}{3} - \frac{1}{2} = \frac{1}{6}$, which is 1 tile. **16.** (a) $\sqrt{200} = 10\sqrt{2}$ m and $\sqrt{50} = 5\sqrt{2}$ m (b) 2 times longer (c) rational — the $\sqrt{2}$ factors divide out: $\frac{10\sqrt{2}}{5\sqrt{2}} = 2$. **17.** $72 = 2^3 \times 3^2$, $40 = 2^3 \times 5$. (a) 2 (giving $144 = 12^2$) (b) 3 (giving $216 = 6^3$) (c) 10 (giving $400 = 20^2$). **18.** (a) $49 < n < 64$, so $n = 50, 55, 60$ (b) $64 < n < 125$, so $n = 72, 81, 90, 99, 108, 117$. **19.** (a) $a = 6^2 \times 2 = 72$ (b) $b = 2^3 \times 5 = 40$ (c) every exponent (6, 4, 2) is even, so the number is a perfect square; halve each exponent to get the square root, $2^3 \times 3^2 \times 5 = 360$.